



SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme : B. tech

Course Code : CSA43

Course Name : Internet programming

Problem Statement (Assignment)

Max Marks:

100

Field	Content
Assignment Title	Smart Campus Service Management System
Course Outcome(s) – CO	CO1: Develop visually appealing web pages using CSS for layout and style, and create dynamic, interactive functionality with JavaScript. CO2: Build dynamic web applications using XML, XSLT, and JSP within the MVC paradigm for efficient data handling and presentation.
Bloom's Taxonomy Level	L3 – Apply L4 – Analyze L5 – Evaluate L6 – Create
SDG Mapping (SDG 1–17)	SDG 9 - Industry, Innovation and Infrastructure, SDG 4 - Quality Education

Problem Statement

A university currently relies on manual communication such as emails, phone calls, and paper-based records for reporting and resolving campus service issues. This approach makes it difficult to efficiently record complaints, assign requests to the appropriate service personnel, monitor progress, and provide timely status updates to students and faculty.

The university therefore proposes to develop a **Smart Campus Service Management System** as an **integrated web-based application**. The system shall enable students and faculty to submit service requests related to laboratory equipment faults, network problems, classroom maintenance, library services, and other technical issues through a web browser. Users should be able to provide relevant details, submit requests, and subsequently track the status of their requests.

The proposed system should demonstrate an **integrated client-server web architecture**. The client side shall provide an interactive and user-friendly interface using HTML/XHTML, CSS, JavaScript, and DOM. The server side shall process requests using Java Servlets and JSP, following the MVC architecture. XML shall be used for structured data representation, and XSLT shall be used to transform XML data for presentation where appropriate.

Problem Identification (10 Marks)-The university relies on manual communication (emails, phone calls, paper records) for reporting and resolving service issues, which causes inefficiency in recording complaints, assigning requests, monitoring progress, and providing timely updates.

Proposed Solution -Develop a **Smart Campus Service Management System (10 Marks)**, a web-based application that allows students and faculty to submit service requests (lab equipment faults, network problems, classroom maintenance, library services) and track their status online.

System Design & Architecture - Data Representation (20 Marks) -Implement an integrated **client-server web architecture**. The client side should use HTML/XHTML, CSS, JavaScript, and DOM for interactivity, while the server side should use Java Servlets and JSP following the **MVC model**. Use **XML** for structured data representation for transforming XML data into user- friendly formats for presentation and reporting.

Implementation & Testing (10 Marks) -Demonstrate request submission, assignment to service personnel, progress tracking, and timely updates. Validate inputs, handle errors, and ensure smooth user experience.

STG Goals –SDG 9 - Industry, Innovation and Infrastructure, **SDG 4 - Quality Education**

Requirement Analysis (SDG 9 – Industry, Innovation & Infrastructure) Identify inefficiencies in manual reporting and propose a digital infrastructure for service management.

Client Interface Development (SDG 4 – Quality Education) Build user-friendly web forms and dashboards so students/faculty can easily submit and track requests, improving access to campus services.

Assessment Criterion	CO Mapping	Weightage / Marks
Problem understanding & formulation	CO1	10
Application of course/domain knowledge	CO2	20
Solution methodology / design / implementation	CO3	20
Use of appropriate modern tools / techniques	CO4	10
Results, testing & validation	CO4 / CO5	15
Analysis, trade-offs & justification	CO5	15
Broader considerations / professional responsibility, where applicable	CO5	5
Technical documentation & reflection	CO1–CO5	5
TOTAL		100

Performance Level	CO Attainment Percentage	Interpretation
Level 3	$\geq 70\%$	CO Attained
Level 2	60–69%	CO Moderately Attained
Level 1	50–59%	CO Minimally Attained
Level 0	$< 50\%$	CO Not Attained

Code of Conduct

I VARSHA S 192462003 (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic-integrity rules will result in disciplinary action.

Signature of the Student:

Varsha

_____ Faculty In-charge Name: Date:	_____ Course Coordinator Name: Date:	_____ Head of Department Name: Date:
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Smart Campus
Service Management System

Integrated Campus Service Management

A unified digital platform for submitting, tracking, and managing campus service requests — replacing emails, phone calls, and paper records.



Fast Request Submission
Submit service requests in under 2 minutes



Real-Time Tracking
Monitor request status with live updates



Digital Service Management
Full lifecycle from submission to resolution

SC

Welcome Back
Sign in to your campus portal

UNIVERSITY EMAIL

yourname@university.edu

PASSWORD

Enter your password

SIGN IN AS

Student

☐ Remember me

[Forgot password?](#)

Sign In to Portal

Quick Demo Access

SC

Smart Campus

Service Management

STUDENT PORTAL

MENU

Dashboard

+ Submit Request

≡ My Requests

🕒 Track Request

🔔 Notifications

🔗 XML / XSLT View

AJ

Alex Johnson

Student

Logout

Good Evening, Alex

Monday, August 31, 2026

Search requests...

🔔

AJ

Dashboard

Overview of your service requests

+ Submit New Request

12

Total Requests

Academic Year 2026

4

Pending

3

In Progress

5

Completed

📌

REQ-1001 has been assigned to IT Support and is currently In Progress. Expected resolution by today, 4:30 PM.

View

Recent Service Requests

View All

Quick Actions

+
Submit Request

🕒
Track Request

≡
My Requests

🔔
Notifications

Recent Activity

●

REQ-1001 — Technician started investigation
11:30 AM

●

REQ-1001 — Assigned to IT Support
10:00 AM

●

REQ-1002 — Request submitted successfully
09:15 AM

●

●

Submit a Service Request

Fill in the details below to report a campus service issue

REQUEST TITLE *

electricity

CATEGORY *

Classroom Maintenance

PRIORITY LEVEL *

Medium

LOCATION / BUILDING *

circular block

ROOM / AREA

304

PROBLEM DESCRIPTION *

fans not working



Request Submitted Successfully

Your service request has been received and will be processed shortly.

Your Request ID

REQ-1009

Save this ID to track your request status

Submit Another

Track Request

REQ-1002

Track

Try: REQ-1001 REQ-1002 REQ-1004 REQ-1003

REQUEST ID

REQ-1002

PENDING

Projector malfunction in Classroom 204

Progress

0% complete

Request Details

CATEGORY

Classroom Maintenance

LOCATION

Academic Block B — Classroom 204

PRIORITY

Medium

STATUS

Pending

ASSIGNED TO

Maintenance Team

SUBMITTED

2026-08-29

Notifications

3 unread notifications

[Mark all as read](#)

All

Unread (3)



Request Assigned REQ-1001

Your request REQ-1001 has been assigned to Arun Kumar — IT Support.

2 hours ago



Status Updated REQ-1002

Request REQ-1002 status has been changed to In Progress by the Maintenance Team.

4 hours ago

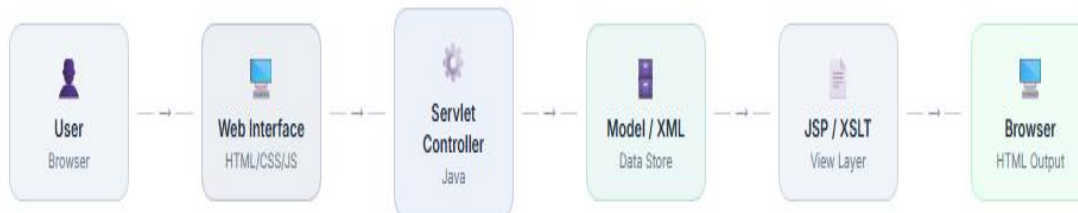


Request Completed REQ-1003

Your request REQ-1003 (Library computer issue) has been successfully completed.

Yesterday

Request Processing Flow



End-to-end request lifecycle from user submission to rendered response

Frontend Technologies

- HTML5 / XHTML
- CSS3 / Tailwind CSS
- JavaScript ES6+
- DOM Manipulation
- Responsive Design

Backend Technologies

- Java Servlet API
- JSP (JavaServer Pages)
- Apache Tomcat Server
- HTTP Request/Response
- Session Management

Data Technologies

- XML Data Storage
- XSD Schema Validation
- XSLT Transformation
- DOM / SAX Parsing
- XPath Querying



SDG 9

Industry, Innovation and Infrastructure

The Smart Campus Service Management System replaces inefficient manual communication — phone calls, emails, and paper records — with an integrated digital infrastructure for campus service requests, assignment, monitoring, and resolution.

HOW OUR SYSTEM CONTRIBUTES:

- ✓ Builds resilient digital infrastructure for campus service management
- ✓ Promotes innovation through automated assignment and real-time tracking
- ✓ Provides inclusive access to service reporting for all campus members
- ✓ Reduces operational overhead through streamlined digital workflows
- ✓ Enables data-driven decision making via analytics dashboards

Target 9.c — Significantly increase access to information and communications technology



SDG 4

Quality Education

The user-friendly digital service portal improves access to campus services and helps students and faculty quickly resolve classroom, laboratory, network, and library issues that would otherwise disrupt learning activities.

HOW OUR SYSTEM CONTRIBUTES:

- ✓ Minimizes disruptions to classroom and laboratory learning environments
- ✓ Ensures network and equipment issues are resolved swiftly
- ✓ Provides equal access to service requests for all students and faculty
- ✓ Maintains quality of campus facilities critical to education
- ✓ Enables transparent communication between users and service teams

Target 4.a — Build inclusive, effective learning environments for all

Projected Impact Metrics

85%

Faster Issue Resolution

100%

Digital Transparency

0

Paper Records Required

24/7

Portal Availability

SC Smart Campus
Service Management

FACULTY PORTAL

MENU

- Dashboard
- + Submit Request
- ≡ My Requests
- ⌚ Track Request
- 🔔 Notifications 3
- 🔗 XML / XSLT View
- 🏠 Architecture

SDG Metrics

PS Prof. Sarah Chen
Faculty

Logout

Good Evening, Prof.
Monday, August 31, 2026

🔍 Search requests...

PS

🔍 Search by ID or title...

All Categories

All Statuses

All Priorities

REQUEST ID	TITLE	CATEGORY	SUBMITTED	PRIORITY	ASSIGNED TO	STATUS	ACTION
REQ-1001	Wi-Fi connectivity problem in Block A Academic Block A	Network Problem	2026-08-28	High	Arun Kumar	In Progress	View Details
REQ-1002	Projector malfunction in Classroom 204 Academic Block B	Classroom Maintenance	2026-08-29	Medium	Maintenance Team	Pending	View Details
REQ-1003	Library computer station issue Central Library	Library Service	2026-08-27	Low	Library IT	Completed	View Details
REQ-1004	Lab equipment malfunction — Centrifuge Unit Science Block	Laboratory Equipment	2026-08-30	Critical	Lab Technician	Assigned	View Details
REQ-1005	Classroom air conditioning not functional Academic Block B	Classroom Maintenance	2026-08-29	Medium	Maintenance Team	In Progress	View Details

Showing 1-5 of 8

⏪ 1 2 ⏩



MVC-Based Architecture

Smart Campus Service Management System

The system implements a classic MVC (Model-View-Controller) pattern with XML as the data store and XSLT for view transformation. Java Servlets act as the controller, coordinating between the XML model and JSP/XSLT views.

System Layer Diagram



CLIENT LAYER

HTML / XHTML

CSS Styling

JavaScript (DOM)

Responsive UI

The presentation layer that users interact with — forms, dashboards, tables, and real-time updates rendered in the browser.

HTTP ↑ XML



CONTROLLER LAYER

Java Servlet

Request Routing

Session Management

Business Logic

Receives and processes HTTP requests, orchestrates data flow between the model and view, and enforces application rules.

HTTP ↑ XML



MODEL / DATA LAYER

XML Data Store

Service Requests

Personnel Records

Schema Validation

Structured XML documents represent all service request data, validated against XSD schemas for data integrity.

HTTP ↑ XML



VIEW LAYER

JSP Templates

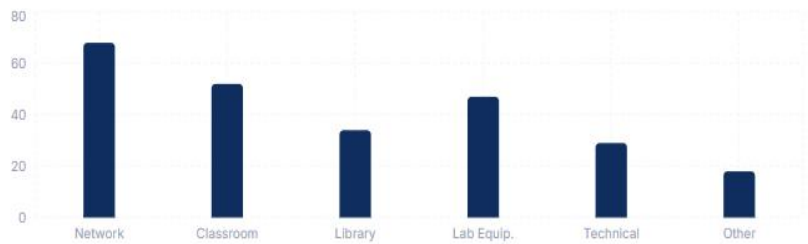
XSLT Transformation

HTML Output

Report Generation

JSP pages and XSLT stylesheets transform raw XML data into human-readable HTML output for the browser.

Requests by Category



Status Distribution



Weekly Request Trend



Analytics & Reports

Campus service performance metrics — Academic Year 2026

[Generate Report](#) [Export XML](#) [XSLT Report](#)

AVG. RESOLUTION TIME

1.8 days

↓ 0.4 days vs last month

RESOLUTION RATE

92.3%

↑ 3.2% vs last month

PENDING > 3 DAYS

12

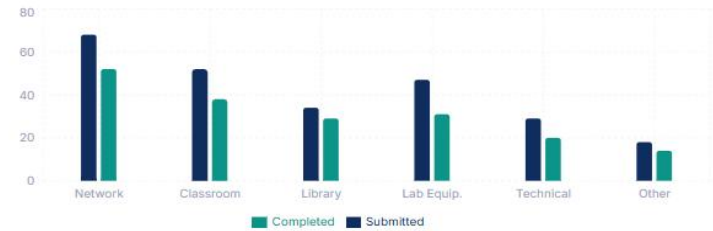
↑ 4 vs last month

USER SATISFACTION

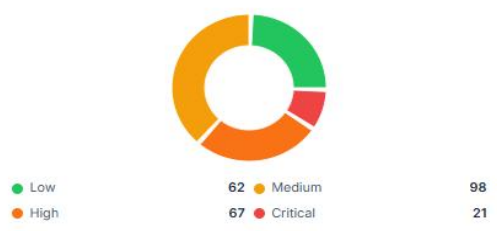
4.2/5

↑ 0.3 vs last month

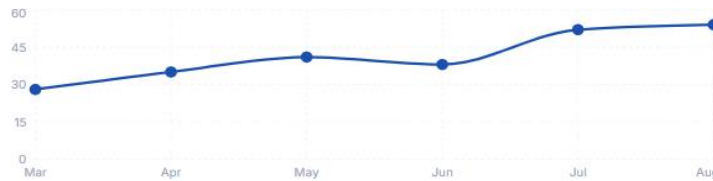
Requests by Category — Submitted vs Completed



Priority Distribution



Monthly Request Volume



Service Quality Radar



Average Resolution Time by Department

